Spotify User Behaviour Analysis

EXECUTIVE SUMMARY

- **Task 1: Subscription Type Distribution**
  - Free Users: **24,903 (~49.81%)**
  - Premium Users: **25,097 (~50.19%)**

Insight:

- Nearly **50-50 split** between free and premium users.
- Indicates a **well-balanced monetisation model**.

- **Task 2: Country-wise User Distribution**
  - Users are **evenly distributed across all countries** in the dataset.
  - No country shows extreme dominance.

Insight:

- Platform has a **uniform global reach**.
- No single geography drives the majority of users.

- **Task 3: Genre Popularity**
  - All genres are **almost equally represented**.
  - No genre significantly dominates others.

Insight:

- User preference is **highly diversified**.
- No dependency on a single content category.

- **Task 4: Churn Analysis**

- Active Users: **~33,333 (~66.67%)**
- Churned Users: **~16,667 (~33.33%)**

Insight:

- One-third of users are inactive → **moderate churn level**.

🔹 **Task 5: Device Usage Analysis**

- Tablet: **9,932 (~19.86%)**
- Mobile: **9,937 (~19.87%)**
- Desktop: **10,037 (~20.07%)**
- Smart Speaker: **10,050 (~20.10%)**
- Car System: **10,044 (~20.09%)**

Insight:

- Device usage is **perfectly balanced (~20% each)**.

🔹 **Task 6: Correlation (Listening Hours vs Skips)**

- Correlation value: **~0 (very weak relationship)**

Insight:

- Listening time and skipping behaviour are **independent variables**.

🔹 **Task 7: Ad Interaction vs Subscription**

- Users interacting with ads are **almost equally split between free & premium**.
- No strong skew observed.

Insight:

- Ad interaction does **not strongly influence subscription upgrade**.

🔹 **Task 8: Listening Hours Distribution**

- Listening hours are **uniformly spread across users**.
- No heavy clustering in a specific range.

Insight:

- No dominant listening behaviour group (e.g., heavy vs light users).

🔹 Task 9: Age Distribution vs Engagement

- Age groups are **evenly distributed**.
- No age group significantly dominates usage.

Insight:

- Platform appeals **equally across all age segments**.

🔹 Task 10: Recommendation Ratings

- Ratings distribution is **even across all rating levels (1–5)**.

Insight:

- User satisfaction is **neutral and evenly spread**.
- No strong positive or negative skew.

Report

(Based on the provided dataset – 50,000 records)

📌 1. Introduction

This project analyses a dataset of **50,000 Spotify users** to understand patterns in:

- Subscription behaviour
- Device usage
- User engagement
- Content preferences
- Churn behaviour

The objective is to extract **data-driven insights without assumptions**, using exploratory data analysis (EDA).

🧹 2. Data Preparation

- Total Records: **50,000**
- Data cleaning performed:
  - Converted categorical values (Yes/No → 1/0)
  - Ensured consistent data types
- No major missing or anomalous values affecting analysis

📈 3. Key Findings & Analysis

🔹 3.1 Subscription Distribution

- Free Users: **24,903 (~49.81%)**
- Premium Users: **25,097 (~50.19%)**

Interpretation:

- The platform shows a **perfectly balanced monetisation structure**

- No skew toward free or paid users

🔹 3.2 Geographic Distribution

- Users are **evenly distributed across all countries**

Interpretation:

- No country dominates the user base
- Dataset reflects **uniform geographic representation**

🔹 3.3 Genre Preferences

- All music genres are **evenly represented**

Interpretation:

- No genre dominates consumption
- Indicates **uniform distribution of content preference**

🔹 3.4 Churn Analysis

- Active Users: **~33,333 (~66.67%)**
- Churned Users: **~16,667 (~33.33%)**

Interpretation:

- One-third of users are inactive
- Churn exists but is **evenly structured within the dataset**

🔹 3.5 Device Usage

- Tablet: **9,932 (~19.86%)**
- Mobile: **9,937 (~19.87%)**
- Desktop: **10,037 (~20.07%)**
- Smart Speaker: **10,050 (~20.10%)**
- Car System: **10,044 (~20.09%)**

Interpretation:

- Device usage is **uniform (~20% each)**
- No platform dependency

🔹 3.6 Listening Behaviour vs Skips

- Correlation: **~0 (no meaningful relationship)**

Interpretation:

- Listening hours and skip behaviour are **independent variables**

🔹 3.7 Ad Interaction & Subscription

- Ad interaction shows **no strong influence on subscription type**

Interpretation:

- Ad engagement does not correlate with premium conversion

🔹 3.8 Listening Hours Distribution

- Listening hours are **evenly distributed across users**

Interpretation:

- No distinct user segmentation (heavy vs light listeners)

🔹 3.9 Age Distribution

- Users are **uniformly distributed across age groups**

Interpretation:

- Platform engagement is **not age-dependent**

🔹 3.10 Recommendation Ratings

- Ratings (1–5) are **evenly distributed**

Interpretation:

- User satisfaction is **neutral and balanced**
- No strong positive or negative trend

🧠 4. Overall Insights

- Every major variable in the dataset shows **uniform distribution**
- No skewness, clustering, or dominance observed
- Relationships between variables are **minimal or non-existent**

⚠️ 5. Critical Observation

The dataset exhibits:

- Near-perfect balance across:
  - Subscription types (~50-50)
  - Devices (~20% each)
  - Ratings (uniform)
  - Demographics (uniform)

Conclusion:

- This indicates a **synthetically generated or evenly distributed dataset**
- Real-world datasets typically show:
  - Skewness
  - Trends
  - Behavioural patterns

🚀 6. Business Implications (Strictly Data-Based)

Since the dataset shows:

- No dominant segment
- No strong correlations

👉 It suggests:

- No single factor drives user behaviour
- All segments contribute equally

🏁 7. Conclusion

This project demonstrates:

- Strong application of **EDA techniques**
- Ability to interpret **balanced datasets**
- Clear identification of:
    - Lack of skewness
    - Absence of strong behavioural patterns